Centre Number	Examination Number

49806119

EXAMINATIONS COUNCIL OF ZAMBIA

Examination for School Certificate Ordinary Level

Design and Technology 6045/1

Paper 1 (Theory & Design)

Tuesday 7 NOVEMBER 2023

Additional Materials
A3 Drawing Paper
Drawing Instruments

Time: 2 hours 30 minutes **Marks: 110**

Instructions to Candidates

1 Write the **centre number** and your **examination number** on **every page** of this question paper and on the **separate Answer Booklet** provided.

2 If working is needed for any question, it must be shown in the spaces provided in the question paper and Answer Booklet.

3 There are **two** sections in this paper, **A** and **B**. Section B has **three** parts **I**, **II** and **III**.
(i) Answer **all** questions in Section A in the spaces provided in this question paper.
(ii) Section B has **three** parts; Part **I** is compulsory.
(iii) Choose **one** question from Part **II** and write your answers in the separate Answer Booklet provided.
(iv) Choose **one** question from Part **III** and use the A3 drawing paper (Insert) provided.
(v) Enter the numbers of questions you have answered from Section B Parts **II** and **III** in the grid provided.

4 Where a freehand sketch is required, it must be drawn to half full size with good linework.

5 At the end of the examination, fasten the Answer Booklet and drawing paper used securely to the question paper.

6 Use your own discretion where measurements are not provided.

Information for Candidates

1 The number of marks is given in brackets [ ] at the end of each question or part question.

2 Use of colour is allowed.

3 All dimensions are in millimetres.

4 Cell phones and calculators are **not allowed** in the examination room.

For Examiner's Use Only

Question	Mark Obtained	Examiner's Initials
Section A		
Section B		
Part I		
Part II		
Part III		
Total		

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SECTION A: [30 MARKS]

Answer all the questions from this section in the spaces provided in this question paper. If working is needed for any question, it must be shown in the space below that question.

1 (a) (i) State **one** risk to personal safety when working in a workroom.

..... [1]

(ii) Mention any precaution that should be taken to a risk stated in **(a)(i)**.

..... [1]

(b) Mention **two** tools you would use for marking a line parallel to a true edge when working with materials.

(ii) ..... [1]

(iii) ..... [1]

(c) **Figure 1** shows the drawing of a symbol of an electronic component.

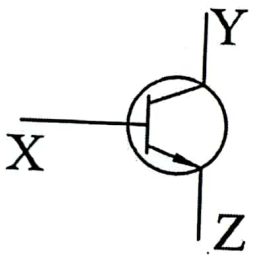


Figure 1

(i) Name the component

Component ..... [1]

(ii) Identify the leg **X**

X ..... [1]

[Total: 6 marks]

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2 **(a)** Two parts of an acrylic stand shown in **Figure 2** are to be glued together.

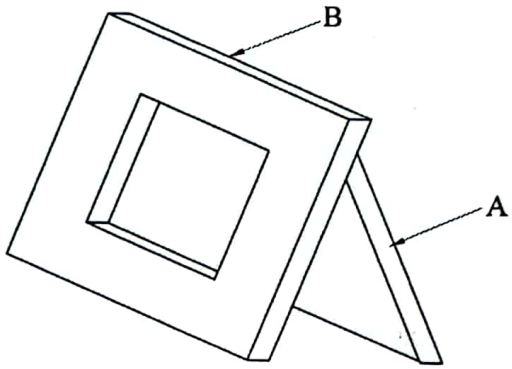


Figure 2

(i) Name the type of glue that could be used for joining part **A** to part **B**.

..... [1]

(ii) State **two** factors that should be considered before joining part **A** to part **B**.

..... [1]

..... [1]

(b) In relation to entrepreneurship, state **three** sources of a business idea.

..... [1]

..... [1]

..... [1]

[Total: 6 marks]

3 **(a)** A drill bit is to be hardened and tempered in the school workroom.

(i) What would happen to a drill bit if it is only hardened?

..... [1]

(ii) Give **one** reason for tempering steels.

..... [1]

(iii) Apart from a drill bit, mention any other tool that can be hardened and tempered.

..... [1]

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(b) **Figure 3** shows the drawing of an electronic circuit diagram.

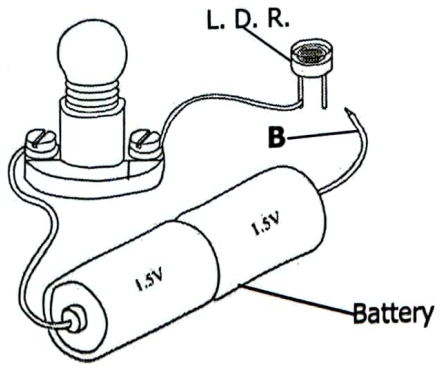


Figure 3

(i) What do the letters L.D.R in the circuit diagram stand for?

..... [1]

(ii) Suggest **one** appropriate method of joining the wire **B** to the L. D. R.

..... [1]

(iii) State the function of a battery in the circuit.

..... [1]

[Total: 6 marks]

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4 (a) **Figure 4** shows the drawing of a hand tool used in the workroom.

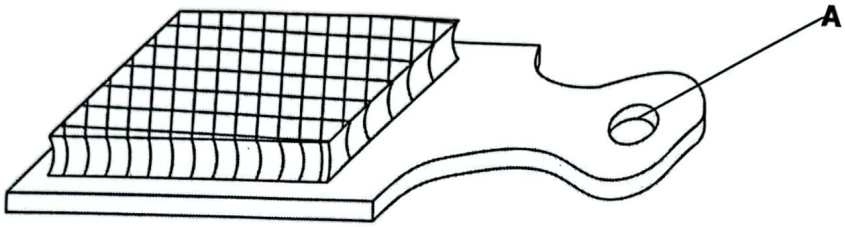


Figure 4

(i) Identify the tool.

.....

[1]

(ii) Mention the function of the tool.

.....

[1]

(iii) State the purpose of hole **A** on the tool.

.....

[1]

(b) Explain the importance of each of the following in a workroom:

(i) Ventilation .....

[1]

(ii) Lighting .....

[1]

(c) Mention **one** recommended method of disposing waste generated in the workroom.

.....

[1]

[Total: 6 marks]

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5 **(a)** **Figure 5** shows a design of a logo for a newly opened school.

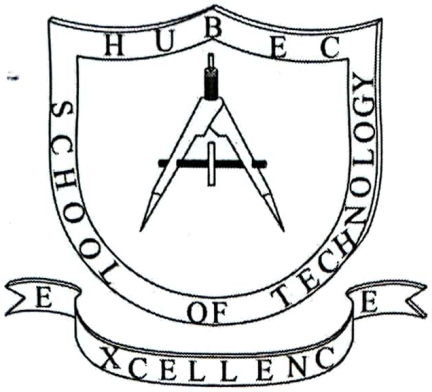


Figure 5

(i) State **two** factors you would consider when designing a logo.

Factor 1 ..... [1]

Factor 2 ..... [1]

(ii) Suggest any suitable material for making the logo.

..... [1]

(b) **Figure 6** shows the drawing of a bench tool.

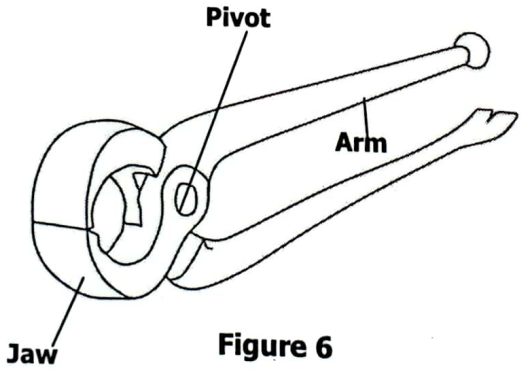


Figure 6

(i) Identify the tool.

..... [1]

(ii) State the class of levers to which the tool belongs.

..... [1]

(iii) Mention the use of the tool in **Figure 6**.

..... [1]

[Total: 6 marks]

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SECTION B: [80 MARKS]

There are three parts in this section; Part I, II and III. Part I is compulsory. Answer one question from Part II and one question from part III. Write your answers in the separate Answer Booklet provided.

Part I: Compulsory

6 Study **Figure 7** and answer the questions that follow.

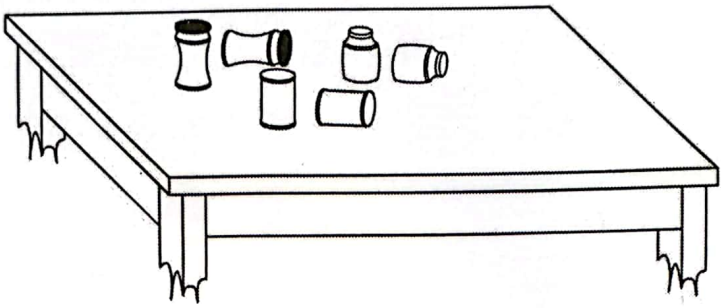


Figure 7

(a) **(i)** Write the **situation** obtaining in **Figure 7**. [2]

(ii) Identify the **problem** from the situation in **a(i)**. [2]

(b) **(i)** Derive a **design brief** from the problem in **a(i)**. [2]

(ii) Using annotations, sketch a pictorial view of one possible solution to the situation. [12]

(c) Mention **two** suitable materials you would use in your solution. [2]

(d) What type of finish would you recommend for your artefact? [1]

(e) Apart from material, suggest **four** specifications you would consider when designing your artefact. [4]

[Total: 25 marks]

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Part II Answer one question

7 (a) **Figure 8** shows a **battery**, a **switch** and a **bell** connected so that the bell rings when the switch is closed.

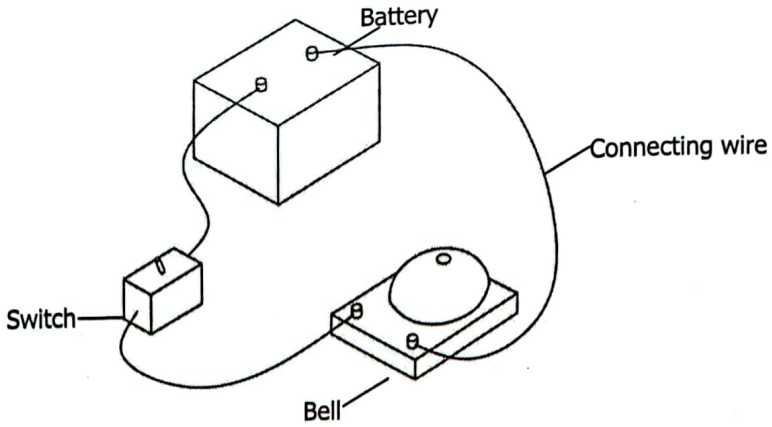


Figure 8

(i) Identify the component which is an **output** in the connection. [1]

(ii) Draw the circuit diagram for this arrangement using standard circuit symbols. [6]

(b) **Figure 9** shows the drawing of one type of a manufactured board.

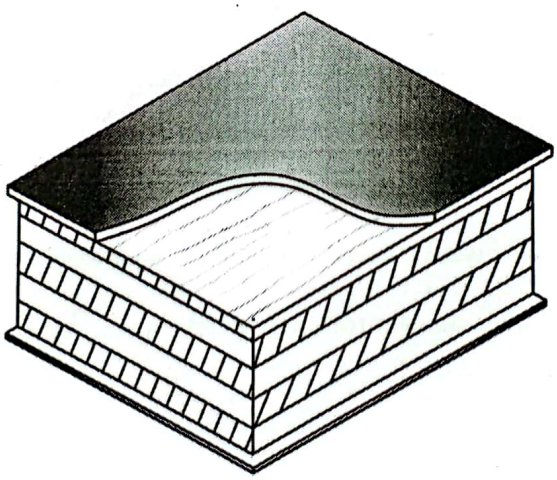


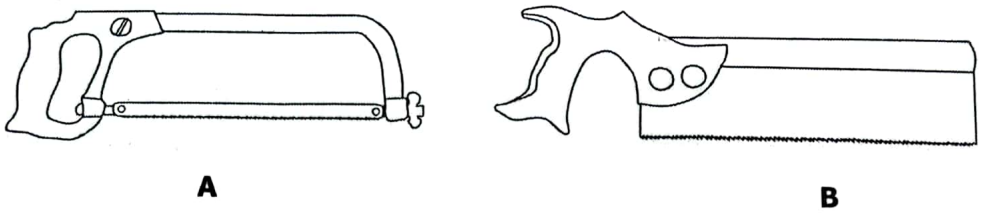
Figure 9

(i) State **two** uses of this manufactured board. [2]

(ii) Explain **two** main advantages of this manufactured board. [2]

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(c) **Figure 10** shows drawings of **two** different types of saws.



A B

Figure 10

(I) Name the saws **A** and **B**. [2]

(II) Suggest **three** factors you would consider when buying a blade for saw **A**. [3]

(d) Most businesses start well and fail after some time. State **three** reasons why businesses fail. [3]

(e) **Figure 11** shows a hinge used in building construction.

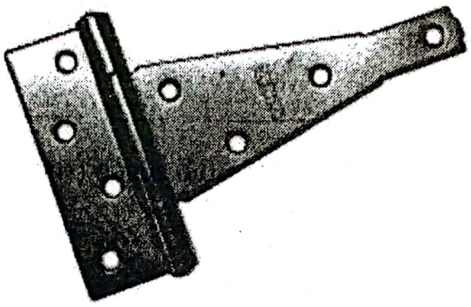


Figure 11

(i) Identify the type of hinge shown. [1]

(ii) Suggest **two** suitable materials for making the hinge. [2]

(iii) Mention **three** areas where the hinge can be applied. [3]

[Total: 25 marks]

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8 **Figure 12** shows a drawing of a kitchen knife.

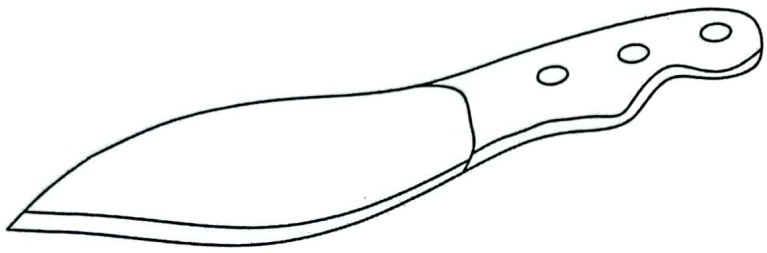


Figure 12

(a) **(i)** Give **two** properties of stainless steel that make it suitable for the blade of the knife . [4]

(ii) State **two** examples of a renewable form of energy source. [2]

(iii) Explain **two** advantages of using the renewable form of energy source. [2]

(b) Tool maintenance and care are very important. Briefly explain the importance of caring for;

(i) Ball pein Hammer, [2]

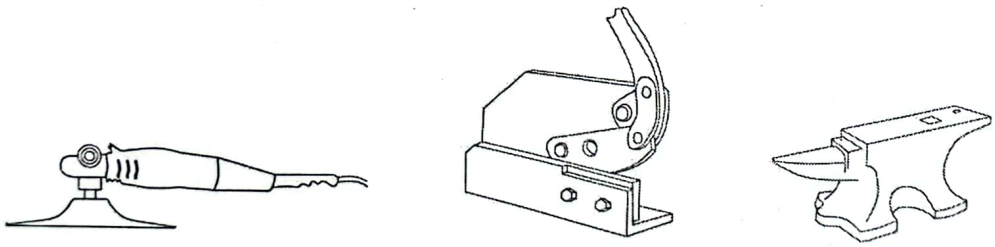
(ii) Chisel. [2]

(c) Orthographic Projection is important in Graphic Communication.

(i) Draw a symbol for First Angle projection. [3]

(ii) Explain the use of a Tee-Square in Graphic Communication. [2]

(d) **Figure 13** shows drawings of machines or equipment that are used when working with metal.



X **Y** **Z**

Figure 13

(i) Explain the function of each tool shown in **Figure 13**. [3]

(ii) Sketch a Sliding Bevel and label any **two** parts. [5]

[Total: 25 marks]

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9 **Figure 14** shows an outdoor workbench.

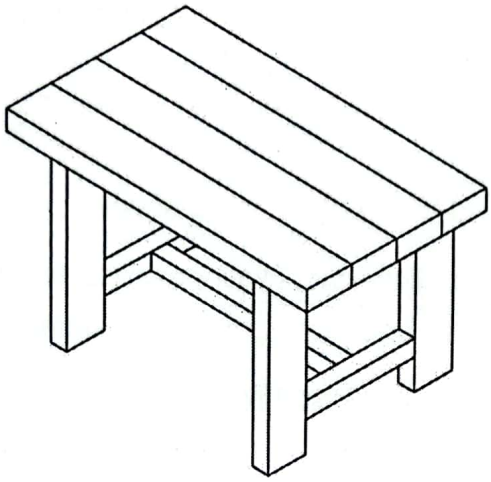


Figure 14

(a) **(i)** Suggest **two** suitable materials for making the workbench. [2]

(ii) Give **two** reasons for each, why the materials in **(a)(i)** are suitable. [4]

(iii) Mention any **two** tools you would use for fixing the top to the frame. [2]

(iv) Using annotated sketches, show how you would fix the top of the bench to the frame. [6]

(b) Sketch the following tools;

(i) Screw driver, [3]

(ii) Rip saw. [3]

(c) Mention **three** personal protective equipment (safety attire) to be worn when welding. [3]

(d) Why is government encouraging people to venture into entrepreneurship? [2]

[Total: 25 marks]

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Part III Answer one question on the A3 drawing paper provided.

10 Deco Day Secondary School has a canteen that serves meals to members of staff during lunch time. You have been asked to design models that would contain a salt seller (cruets) and a bottle of chilli to be placed on each table in the canteen. The containers could be of any shape as long as they are **not** cylinders.

(a) Sketch a model for proposed design in pictorial view. [3]

(b) Construct the surface development of your proposed design model. [12]

(c) Suggest the name of the canteen and print it on your model. [3]

(d) Differentiate using sketches by indicating the container for salt and chili. [12]

[Total: 30 marks]

11 A design of a large lamp shade is to be fixed outside the Home Economics Room.

The shade has the shape of a truncated hexagonal pyramid of A/C = 60. It is made of thin material. A Front Elevation of the lamp shade is given in **Figure 18**.

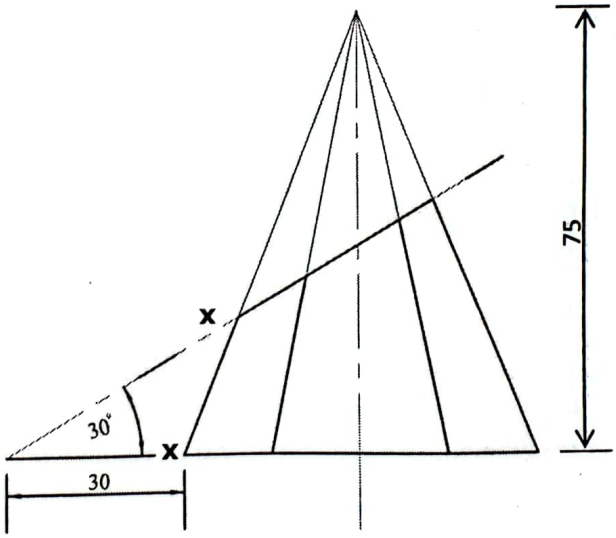


Figure 18

(a) **(i)** Copy the given view. [6]

(ii) Determine the pattern of the lamp shade (surface development) before folding it along **x – x**. [14]

(b) In order to prevent the transmission of cholera, the school wants to ensure that learners abide by hygiene regulations.

Design a sign to illustrate graphically the message 'WASH YOUR HANDS,' in a circle of diameter 80. [10]

[Total: 30 marks]

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